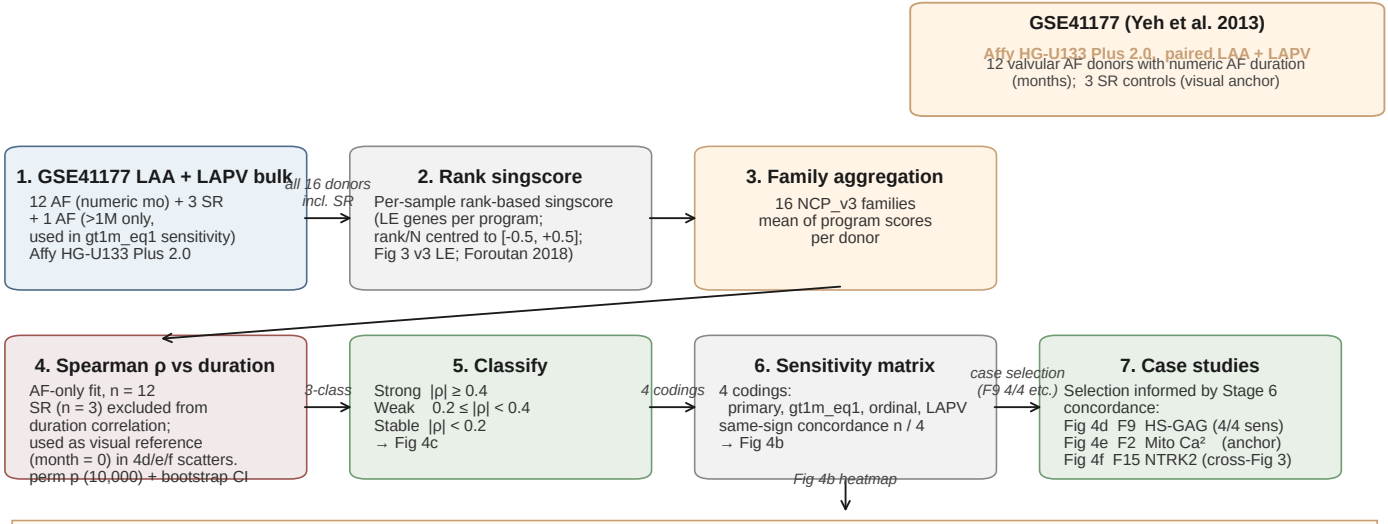
Fig 4a AF-duration chronicity pipeline (LAA + LAPV bulk)



Methodological limitations (acknowledged in Discussion)

(i) GSE41177 LAA AF n = 12 (donors with numeric duration); statistical power to detect |ρ| ≥ 0.4 (Spearman, α = 0.05) is well below 80%. Fig 4 is therefore an effect-size-prioritised exploratory analysis, not a high-powered association screen.

(ii) No family reaches the strict criterion |ρ| ≥ 0.4 + p < 0.05; classification is by effect-size partition only (3-class above).

(iii) GSE41177 is single-cohort valvular AF (no cross-cohort meta-analysis at this duration resolution); LAPV is reported as a region sensitivity arm.

(iv) Fig 4 adopts a 3-class framework parallel to Fig 3 (cellular attribution) on the temporal axis. The two are formally analogous, not biologically equivalent.